

Forecasting for the ARMA(1,1) Process

$$X_t = \phi X_{t-1} + \theta Z_{t-1} + Z_t$$

$$\Rightarrow (1 - \phi B) X_t = (1 + \theta B) Z_t \quad (+)$$

For the k step ahead forecast, having observed the series to $t=n$, we have

~~For~~

$$\hat{X}_{n+k}(n) = \phi \hat{X}_{n+k-1}(n) + \theta [Z_{n+k-1}] + [Z_{n+k}]$$

$$[Z_t] = \begin{cases} Z_t & t \leq n; \\ 0 & t > n. \end{cases}$$

$k=1$

$$\hat{X}_{n+1} - \phi \hat{X}_n = 0 + \theta Z_n$$

$$\Rightarrow \hat{X}_{n+1} = \phi X_n + \theta Z_n$$

$k=2$

$$\hat{X}_{n+2} - \phi \hat{X}_{n+1} = 0$$

$$\hat{X}_{n+2} = \phi \hat{X}_{n+1}$$

$$= \phi [\phi X_n + \theta Z_n]$$

$$= \phi^2 X_n + \phi \theta Z_n$$

Generally : $\hat{X}_{n+k}(n) = \phi^{k-1} [\phi X_n + \theta Z_n]$

「We require Z_n in terms of $\{X_t\}$.」

From (†):

$$Z_t = \frac{(1 - \phi B) X_t}{1 + \theta B}$$

$$= (1 + \theta B)^{-1} (1 - \phi B) X_t$$

$$(1 + \theta B)^{-1} = \sum_{j=0}^{\infty} (-\theta)^j B^j \quad \leftarrow \text{Taylor series expansion}$$

$$\therefore Z_t = (1 - \phi B) \sum_{j=0}^{\infty} (-\theta)^j B^j X_t$$

$$= (1 - \phi B) \sum_{j=0}^{\infty} (-\theta)^j X_{t-j}$$

$$= X_t - (\theta + \phi) \sum_{j=1}^{\infty} (-\theta)^{j-1} X_{t-j}$$

Finally

$$\hat{X}_{n+k}(n) = \phi^{k-1} [\phi X_n + \theta \{X_n - (\theta + \phi) \sum_{j=1}^{\infty} (-\theta)^{j-1} X_{n-j}\}]$$